# Scorecard Update

Erik Larsen

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## Overview

This markdown shows the data logging process of each round, and shows performance trends and metric averages

## Set-Up Environment

### Attach Packages

```
library(golf)
library(tidyverse)
library(DBI)
library(duckdb)
```

## Record New Scorecard

### Input the Scores Data

```
# required inputs to fill out the scorecard

round_course <- 'Randolph North'
round_date <- '2026-08-02'
round_tees <- 'white'

hole_scores <- c(5, 5, 6, 4, 4, 3, 4, 4, 4,
                 5, 4, 5, 6, 5, 3, 5, 4, 6)

FIRs <- c(rep(0, 4), 1, 0, 1, 0, 0,
          rep(0, 9))

GIRs <- c(rep(0, 3), rep(1, 6),
          0, 0, 1, rep(0,3), 1, 0, 0)

putts_rec <- c(1, 1, 2, 2, 2, 2, 2, 3, 1,
               1, 1, 3, 2, 2, 1, 2, 1, 3)

chips_rec <- c(1, 1, 1, rep(0,5), 1,
```

```
               2, 2, 0, 1, 1, 1, 0, 2, 1)

penalties_rec <- c(0, 1, rep(0,16))

tee_clubs <- c('D', 'D', 'D', 'D', 'D', 'SW', 'D', '6', 'D',
               'D', '6', 'D', 'D', 'D', '9', 'D', 'D', 'D')

index <- 9.8
```

### Specify Course and Tees

```
# get the scorecard for the new round, ensuring hole-by-hole scores are filled
if ( length(hole_scores) > 0 ) {
 Card <- golf::get_course(course = round_course, date = round_date, tees = round_tees)
}
```

### Get Scoring Metrics

```
# fill the scorecard for the new round, ensuring hole-by-hole scores are filled
if ( length(hole_scores) > 0 ) {
 Card <- golf::log_score(Scorecard = Card,
                       hole_by_hole = hole_scores,
                       name = 'Erik Larsen',
                       index = index,
                       FIR = FIRs,
                       GIR = GIRs,
                       putts = putts_rec,
                       chips = chips_rec,
                       penalties = penalties_rec,
                       tee_club = tee_clubs)
}
```

## Update the db

### Update the Players Table

```
con <- golf::get_db_connection()

# if the logged data is newer than the last logged round in the 'rounds' table,
# update the 'players' table

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DES
     dplyr::distinct(date) |>
     unlist() %>%
     lubridate::as_date(.) |>
     as.character() < round_date &

     length(hole_scores) > 0
```

```
    ) {

  DBI::dbAppendTable(conn = con,
                     name = 'players',
                     value = players <- Card |>
                       dplyr::select(player_id, player_name, GHIN, index, date) |>
                       dplyr::distinct() |>
                       dplyr::rename(handicap_index = index) |>
                       dplyr::mutate(date = as.character(date))
                     )

}
```

#### Update the Courses Table

```
con <- golf::get_db_connection()

# if the logged data is newer than the last logged round in the 'rounds' table,
# update the 'courses' table

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DESC
     dplyr::distinct(date) |>
     unlist() %>%
     lubridate::as_date(.) |>
     as.character() < round_date &

     length(hole_scores) > 0
     ) {
  DBI::dbAppendTable(conn = con,
                     name = 'courses',
                     value = course <- Card |>
                       dplyr::select(course, tees, to_par, slope, course_rating, hole, yds, par, hole_hand
                       dplyr::distinct() |>
                       dplyr::rename(course_name = course) |>
                       dplyr::mutate(hole = gsub(hole,
                                                 pattern = 'hole_',
                                                 replacement = '') |> as.numeric())
                     )
}
```

#### Update the Rounds Table

```
con <- golf::get_db_connection()

# if the logged data is newer than the last logged round in the 'rounds' table,
# update the 'rounds' table

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DESC
     dplyr::distinct(date) |>
     unlist() %>%
```

```r
    lubridate::as_date(.) |>
    as.character() < round_date &

    length(hole_scores) > 0
    ) {
  DBI::dbAppendTable(conn = con,
                     name = 'rounds',
                     value = Card |>
                       dplyr::select(-to_par, -slope, -course_rating, -yds, -par) |>
                       dplyr::distinct() |>
                       dplyr::rename(handicap_index = index) |>
                       dplyr::rename(course_name = course) |>
                       dplyr::mutate(date = as.character(date))  |>
                       dplyr::mutate(hole = gsub(hole,
                                                 pattern = 'hole_',
                                                 replacement = '') |> as.numeric()) |>
                       dplyr::group_by(course_name, date) |>
                       dplyr::arrange(hole) |>
                       dplyr::ungroup() |>
                       dplyr::relocate(c(tot_gross, tot_net), .after = IN_net) |>
                       dplyr::relocate(course_handicap, .after = tees)
                     )
}
```

#### Get Club Metrics df

```r
# create a dataframe, specifying the shape for tracked shots on each hole
  # -> all strokes from off the green

club_metrics_df <- golf::get_tracked_shots_data_shape(round_date = round_date) |>
  dplyr::arrange(hole)
```

#### Annotate Club Metrics

```r
# manually annotate from Garmin Golf App log
club_choice <- c(
  'D', 'GW', 'SW', 'GW',
  'D', 'SW', 'P',
  'D', '5', '7', 'PW',
  'D', '9',
  'D', 'SW',
  'SW',
  'D', 'SW',
  '6',
  'D', 'PW', 'GW',

  'D', '4', 'SW', 'PW',
  '6', 'PW', 'PW',
  'D', '8',
  'D', '4', '9', 'PW',
```

```r
  'D', '5', 'LW',
  '9', 'GW',
  'D', '5', '8',
  'D', '7', 'LW',
  'D', 'PW', 'LW'
)

dist_to_target <- c(
  270, 131, 98, 19,
  270, 112, 11,
  270, 50, 170, 20,
  270, 153,
  270, 93,
  83,
  270, 87,
  210,
  270, 166, 23,

  270, 140, 18, 13,
  191, 50, 15,
  270, 163,
  270, 220, 152, 23,
  270, 100, 30,
  173, 26,
  270, 210, 160,
  270, 65, 20,
  270, 141, 27
)

yds <- c(
  228, 34, 85, 23,
  263, 114, 12,
  297, 57, 191, 26,
  267, 150,
  325, 96,
  92,
  285, 87,
  192,
  268, 153, 21,

  284, 122, 5, 20,
  221, 36, 14,
  206, 171,
  266, 114, 130, 18,
  287, 113, 30,
  164, 31,
  261, 121, 160,
  320, 79, 20,
  265, 165, 20
)
lie_type <- c(
  'tee', 'sand', 'rough', 'fairway',
  'tee', 'fairway', 'fairway',
```

```r
  'tee', 'rough', 'fairway', 'fairway',
  'tee', 'rough',
  'tee', 'fairway',
  'tee',
  'tee', 'fairway',
  'tee',
  'tee', 'fairway', 'rough',

  'tee', 'rough', 'fairway', 'fairway',
  'tee', 'rough', 'fairway',
  'tee', 'rough',
  'tee', 'rough', 'rough', 'fairway',
  'tee', 'rough', 'rough',
  'tee', 'rough',
  'tee', 'rough', 'fairway',
  'tee', 'rough', 'sand',
  'tee', 'rough', 'sand'
)

target_status <- c(
  'no', 'no', 'yes', 'yes',
  'no', 'no', 'yes',
  'no', 'yes', 'no', 'yes',
  'no', 'yes',
  'yes', 'yes',
  'yes',
  'yes', 'yes',
  'yes',
  'no', 'yes', 'yes',

  'no', 'yes', 'no', 'yes',
  'no', 'no', 'yes',
  'no', 'yes',
  'no', 'no', 'no', 'yes',
  'no', 'no', 'yes',
  'no', 'yes',
  'no', 'no', 'yes',
  'no', 'no', 'yes',
  'no', 'no', 'yes'
)

location <- c(
  'left', 'short', 'on_target', 'on_target',
  'left', 'left', 'on_target',
  'left', 'on_target', 'long', 'on_target',
  'right', 'on_target',
  'on_target', 'on_target',
  'on_target',
  'on_target', 'on_target',
  'on_target',
  'right', 'on_target', 'on_target',

  'right', 'on_target', 'short', 'on_target',
```

```r
  'long', 'short', 'on_target',
  'left', 'on_target',
  'right', 'short', 'short', 'on_target',
  'right', 'long', 'on_target',
  'right', 'on_target',
  'right', 'short', 'on_target',
  'long', 'long', 'on_target',
  'right', 'long', 'on_target'
)

type_of_shot <- c(
  'tee', 'fwbunker', 'punch', 'chip',
  'tee', 'full', 'chip',
  'tee', 'punch', 'full', 'chip',
  'tee', 'full',
  'tee', 'choked',
  'tee',
  'tee', 'choked',
  'tee',
  'tee', 'full', 'chip',

  'tee', 'punch', 'chip', 'chip',
  'tee', 'chip', 'chip',
  'tee', 'full',
  'tee', 'full', 'full', 'chip',
  'tee', 'punch', 'chip',
  'tee', 'chip',
  'tee', 'full', 'full',
  'tee', 'punch', 'gsbunker',
  'tee', 'full', 'gsbunker'
)
```

#### Get Shot Metrics

```r
con <- golf::get_db_connection()

# if the data is newer than the last round in the 'club_metrics' table, append
  # the new annotations (expands from 18 rows by the number determined from the
  # get_tracked_shots_data_shape function)

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM club_metrics ORDER BY da
     dplyr::distinct(date) |>
     unlist() %>%
     lubridate::as_date(.) |>
     as.character() < round_date &

     length(hole_scores) > 0
) {
  length(club_choice)
  club_metrics <- golf::harmonize_club_metrics(club_metrics = club_metrics_df,
                                               club_choice = club_choice,
                                               distance_to_target = dist_to_target,
```

```
                                                    distance_traveled = yds,
                                                    lie_type = lie_type,
                                                    target_status = target_status,
                                                    location = location,
                                                    type_of_shot = type_of_shot
                                                    )
  club_metrics <- club_metrics |>
    dplyr::mutate(dplyr::across(c(dplyr::contains("yds")), ~as.numeric(.x)))
  head(club_metrics)
}

club_metrics_df |> dplyr::select(tracked_shots) |> dplyr::mutate(sum(tracked_shots))
```

```
##    tracked_shots sum(tracked_shots)
## 1              4                 49
## 2              3                 49
## 3              4                 49
## 4              2                 49
## 5              2                 49
## 6              1                 49
## 7              2                 49
## 8              1                 49
## 9              3                 49
## 10             4                 49
## 11             3                 49
## 12             2                 49
## 13             4                 49
## 14             3                 49
## 15             2                 49
## 16             3                 49
## 17             3                 49
## 18             3                 49
```

#### Update the Club Metrics Table

```
con <- golf::get_db_connection()

# if the data is newer than the last round in the 'club_metrics' table, append
  # the new data to the table

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM club_metrics ORDER BY da
     dplyr::distinct(date) |>
     unlist() %>%
     lubridate::as_date(.) |>
     as.character() < round_date &

     length(hole_scores) > 0
     ) {
  DBI::dbAppendTable(conn = con,
                     name = 'club_metrics',
                     value = club_metrics |>
```

```
                      dplyr::mutate(date = as.character(date))
                    )
}
```

```
# clean the global environment
rm(list = ls()[which(grepl(ls(), pattern= 'con|round_course|round_date|index|real_db')==F)])
```

## Summarize Metrics

### Gather and Format

Gather and format from the database

```
con <- golf::get_db_connection()

# get the hole-by-hole scores from each round in the database

  # join the hole-by-hole scores from the 'rounds' table to the 'courses' table
    # need each hole's par rating
    # need each course's course_rating from the 'courses' table

  # join the 'rounds' and 'courses' tables to the 'players' table
    # need handicap_index from the 'players' table

scores <- DBI::dbGetQuery(conn = con, statement = paste0(
  "SELECT DISTINCT r.*, c.par, c.course_rating FROM rounds r
  INNER JOIN courses c
  ON c.tees = r.tees
  AND c.course_name = r.course_name
  AND c.hole = r.hole
  INNER JOIN players p
  ON r.GHIN = p.GHIN
  AND r.handicap_index = p.handicap_index
  AND r.date = p.date;"
)) |>
  dplyr::mutate(date = lubridate::as_date(date)) |> # convert strings to date's
  dplyr::relocate(par, .after = hole) |>
  dplyr::relocate(course_rating, .after = tees) |>
  dplyr::group_by(date) |>
  dplyr::arrange(desc(date), hole) |>
  dplyr::ungroup()
```

```
con <- golf::get_db_connection()

# get stroke metrics from the 'club_metrics' table

stroke_quality <- DBI::dbGetQuery(conn = con, statement = paste0(
  "SELECT DISTINCT * FROM club_metrics;"
)) |>
  dplyr::mutate(date = lubridate::as_date(date)) |> # convert strings to date's
  dplyr::group_by(date) |>
```

```
  dplyr::arrange(desc(date), hole, stroke) |>
  dplyr::ungroup()
```

### Compute Metrics

Compute standard metrics

```
# define metrics
scores_sum <- scores |>
  dplyr::mutate(date_course = paste0(date,
                                    '\n',
                                    course_name, '\n',
                                    handicap_index),
                chips = dplyr::case_when(
                  grepl(date, pattern = '07-13') ~ NA_real_, TRUE ~ chips),
                `chips+putts` = chips+putts,
                FIR_opps = dplyr::case_when(par > 3 ~ 1, TRUE ~ NA),
                `Iron FIRs` = dplyr::case_when(par > 3 &
                                                grepl(tee_club, pattern = '4|5') &
                                                FIR == 1 ~ 1,
                                              TRUE ~ 0),
                `Iron FIR opps` = dplyr::case_when(par > 3 &
                                                    grepl(tee_club, pattern = '4|5') ~ 1,
                                                  TRUE ~ 0),
                `Driver FIRs` = dplyr::case_when(par > 3 &
                                                  tee_club == 'D' &
                                                  FIR == 1 ~ 1,
                                                TRUE ~ 0),
                `Driver FIR opps` = dplyr::case_when(par > 3 &
                                                      tee_club == 'D' ~ 1,
                                                    TRUE ~ 0),
                `Par 3 GIRs` = dplyr::case_when(par == 3 &
                                                 GIR == 1 ~ 1,
                                               TRUE ~ 0),
                greenie_putts = dplyr::case_when(GIR == 1 ~ putts, TRUE ~ NA_real_),
                updown_conv = dplyr::case_when(GIR == 0 &
                                                par == gross ~ 1,
                                              TRUE ~ 0),
                updown_opps = dplyr::case_when(GIR == 0 &
                                                chips > 0 ~ 1,
                                              TRUE ~ 0)
                ) |>
  dplyr::rename(`Handicap Index` = handicap_index)

# summarize by round
scores_sum <- scores_sum |>
  dplyr::group_by(date, date_course, course_rating, `Handicap Index`) |>
  dplyr::summarize(
                  FIRs = sum(FIR, na.rm = F),
                  `Iron FIRs` = sum(`Iron FIRs`, na.rm = T),
                  `Iron FIR%` = round(((sum(`Iron FIRs`, na.rm = T)/sum(`Iron FIR opps`, na.rm = T))*1(
                  `Driver FIRs` = sum(`Driver FIRs`, na.rm = T),
                  `Driver FIR%` = round(((sum(`Driver FIRs`, na.rm = T)/sum(`Driver FIR opps`, na.rm =
```

```r
                   `FIR%` = round((sum(FIRs, na.rm = T)/sum(FIR_opps, na.rm = T))*100, 1),
                   GIRs = sum(GIR, na.rm = F),
                   `Par 3 GIRs` = sum(`Par 3 GIRs`, na.rm = T),
                   `GIR%` = round((sum(GIRs, na.rm = T)/18)*100, 1),
                   putts = sum(putts, na.rm = F),
                   `Avg GIR putts` = round(mean(greenie_putts, na.rm = T), 2),
                   chips = sum(chips, na.rm = F),
                   `chips+putts` = sum(`chips+putts`, na.rm = F),
                   `UpDown%` = round(((sum(updown_conv, na.rm = T)/sum(updown_opps, na.rm = T))*100), 1
                   pars = sum(is_gross_par, na.rm = F),
                   birdies = sum(is_gross_birdie, na.rm = F),
                   bogies = sum(is_gross_bogey, na.rm = F),
                   `doubles+` = sum(is_gross_bogey_worse, na.rm = F),
                   penalties = sum(penalties, na.rm = T),
                   `Gross Score` = mean(tot_gross, na.rm = F),
                   `Net Score` = mean(tot_net, na.rm = F)) %>%
  dplyr::mutate(dplyr::across(c(`Iron FIRs`, `Driver FIRs`,
                                `Iron FIR%`, `Driver FIR%`, `FIR%`),
                              ~dplyr::if_else(is.na(FIRs), NA, .)),

                dplyr::across(c(`Par 3 GIRs`, `Avg GIR putts`,
                                `UpDown%`, `GIR%`),
                              ~dplyr::if_else(is.na(GIRs), NA, .)),

                `chips+putts` = dplyr::case_when(is.na(chips) ~ NA_real_,
                                                 TRUE ~ `chips+putts`)) %>%

  dplyr::mutate(

    `UpAndDown%` = dplyr::case_when(

      grepl(date, pattern = '07-13|09-21') ~ NA, TRUE ~ `UpDown%`),

    `Iron FIR%` = dplyr::case_when(`Iron FIR%` == NaN ~ 0.0, TRUE ~ `Iron FIR%`))
```

```r
head(scores_sum |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 26
## # Groups:   date, date_course, course_rating [6]
##   date       date_course       course_rating `Handicap Index`  FIRs `Iron FIRs` `Iron FIR%` `Driver
##   <date>     <chr>                     <dbl>            <dbl> <int>       <dbl>       <dbl>
## 1 2026-08-02 "2026-08-02\nRand~         69.8              9.8     2           0         NaN
## 2 2026-07-31 "2026-07-31\nSilv~         68.9              9.8     5           0         NaN
## 3 2026-07-26 "2026-07-26\nRand~         69.8             10.2     3           0         NaN
## 4 2026-07-12 "2026-07-12\nDell~         67.8             10.2     5           1         100
## 5 2026-07-05 "2026-07-05\nRand~         69.8             10.5     9           0         NaN
## 6 2026-07-03 "2026-07-03\nRand~         69.8             10.6     1           0         NaN
## # i 9 more variables: `UpDown%` <dbl>, pars <int>, birdies <int>, bogies <int>, `doubles+` <int>, pe
```

### View Metrics

Separate and view metrics

**Scoring Metrics** Scores and Handicap

```
scoring_metrics <- scores_sum |>
  dplyr::select(`Handicap Index`, `Gross Score`, `Net Score`)
head(scoring_metrics |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 6
## # Groups:   date, date_course, course_rating [6]
##   date       date_course                           course_rating `Handicap Index` `Gross Score` `Net Sc
##   <date>     <chr>                                         <dbl>            <dbl>         <dbl>       <
## 1 2026-08-02 "2026-08-02\nRandolph North\n9.8"              69.8              9.8            82
## 2 2026-07-31 "2026-07-31\nSilverbell\n9.8"                  68.9              9.8            82
## 3 2026-07-26 "2026-07-26\nRandolph North\n10.2"             69.8             10.2            80
## 4 2026-07-12 "2026-07-12\nDell Urich\n10.2"                 67.8             10.2            85
## 5 2026-07-05 "2026-07-05\nRandolph North\n10.5"             69.8             10.5            78
## 6 2026-07-03 "2026-07-03\nRandolph North\n10.6"             69.8             10.6            81
```

**Stroke Metrics** In golf, every hole has a `par`– an average number of strokes taken to get the ball in the hole

There are (almost always), three different `par`s on every course:

- `par` 3s
- `par` 4s
- `par` 5s

A shorthand to determine how a golfer performed across holes is to rate how many strokes *relative to par* they were (-1, -2, +1, +2, etc)

These numbers are given names:

- `0 = par`
- `-2 = eagle`
- `-1 = birdie`
- `+1 = bogey`
- `+2 = double bogey`

I quantify these below:

```
stroke_metrics <- scores_sum |>
  dplyr::select(`doubles+`, bogies, pars, birdies)
head(stroke_metrics |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 7
## # Groups:   date, date_course, course_rating [6]
##   date       date_course                           course_rating `doubles+` bogies  pars birdies
##   <date>     <chr>                                         <dbl>      <int>  <int> <int>   <int>
## 1 2026-08-02 "2026-08-02\nRandolph North\n9.8"              69.8          1      9     7       1
## 2 2026-07-31 "2026-07-31\nSilverbell\n9.8"                  68.9          1     10     7       0
## 3 2026-07-26 "2026-07-26\nRandolph North\n10.2"             69.8          2      5     9       2
## 4 2026-07-12 "2026-07-12\nDell Urich\n10.2"                 67.8          2     10     6       0
## 5 2026-07-05 "2026-07-05\nRandolph North\n10.5"             69.8          2      4    10       2
## 6 2026-07-03 "2026-07-03\nRandolph North\n10.6"             69.8          2      6     9       1
```

**Around-the-Green Metrics** `Chips`: + strokes around the green taken to get onto the green

`Putts`: + strokes taken with the putter on the green

`Avg GIR putts`: + Average # of putts on holes where the ball was hit on to the green within 2 strokes of par (green in regulation, `GIR`)

`UpDown%` (aka `Scramble%`): + # of holes without a `GIR` but par was made / # of holes without a `GIR`

```r
atg_metrics <- scores_sum |>
  dplyr::select(chips, `chips+putts`, `UpAndDown%`, putts, `Avg GIR putts`)
head(atg_metrics |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 8
## # Groups:   date, date_course, course_rating [6]
##   date       date_course                          course_rating chips `chips+putts` `UpAndDown%` putts
##   <date>     <chr>                                        <dbl> <dbl>         <dbl>        <dbl> <int>
## 1 2026-08-02 "2026-08-02\nRandolph North\n9.8"             69.8    14            46         20      32
## 2 2026-07-31 "2026-07-31\nSilverbell\n9.8"                 68.9    18            49         21.4    31
## 3 2026-07-26 "2026-07-26\nRandolph North\n10.2"            69.8    17            47         40      30
## 4 2026-07-12 "2026-07-12\nDell Urich\n10.2"                67.8    15            52         20      37
## 5 2026-07-05 "2026-07-05\nRandolph North\n10.5"            69.8    12            42         62.5    30
## 6 2026-07-03 "2026-07-03\nRandolph North\n10.6"            69.8    15            44         41.7    29
```

**Ball Striking Metrics** Approach and tee accuracy

`GIR (green in regulation)`: + hole where the ball was hit on to the green within 2 strokes of par

`FIR (fairway in regulation)`: + hole where the ball was hit on to the fairway from the tee box (only on par 4's and par 5's)

`Iron FIR`: + hole where an iron was used off the tee for a `FIR`

`Driver FIR`: + hole where driver was used off the tee for a `FIR`

```r
ball_striking_metrics <- scores_sum |>
  dplyr::select(GIRs, `GIR%`, `Par 3 GIRs`,
                FIRs, `FIR%`, `Iron FIRs`, `Iron FIR%`,
                `Driver FIRs`, `Driver FIR%`)
head(ball_striking_metrics |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 12
## # Groups:   date, date_course, course_rating [6]
##   date       date_course                          course_rating  GIRs `GIR%` `Par 3 GIRs`  FIRs `FIR%`
##   <date>     <chr>                                        <dbl> <int>  <dbl>        <dbl> <int>  <dbl>
## 1 2026-08-02 "2026-08-02\nRandolph North\n9.8"             69.8     8   44.4            2     2   14.3
## 2 2026-07-31 "2026-07-31\nSilverbell\n9.8"                 68.9     4   22.2            2     5   38.5
## 3 2026-07-26 "2026-07-26\nRandolph North\n10.2"            69.8     7   38.9            1     3   21.4
## 4 2026-07-12 "2026-07-12\nDell Urich\n10.2"                67.8     6   33.3            2     5   38.5
## 5 2026-07-05 "2026-07-05\nRandolph North\n10.5"            69.8     9   50              1     9   64.3
## 6 2026-07-03 "2026-07-03\nRandolph North\n10.6"            69.8     5   27.8            1     1    7.1
```

**Club Metrics** Yardage and accuracy for each club

```
## # A tibble: 6 x 6
## # Groups:   date [1]
##   date       club      n rd_min_yds_to_target rd_min_yds_traveled rd_min_yd_diff
##   <date>     <chr> <int>                <dbl>               <dbl>          <dbl>
## 1 2026-08-02 4         2                  140                 114             18
## 2 2026-08-02 5         3                   50                  57            -13
## 3 2026-08-02 6         2                  191                 192            -30
## 4 2026-08-02 7         2                   65                  79            -21
## 5 2026-08-02 8         2                  160                 160             -8
## 6 2026-08-02 9         3                  152                 130              3
```

```
## # A tibble: 6 x 6
## # Groups:   date [1]
##   date       club      n rd_max_yds_to_target rd_max_yds_traveled rd_max_yd_diff
##   <date>     <chr> <int>                <dbl>               <dbl>          <dbl>
## 1 2026-08-02 4         2                  220                 122            106
## 2 2026-08-02 5         3                  210                 121             89
## 3 2026-08-02 6         2                  210                 221             18
## 4 2026-08-02 7         2                  170                 191            -14
## 5 2026-08-02 8         2                  163                 171              0
## 6 2026-08-02 9         3                  173                 164             22
```

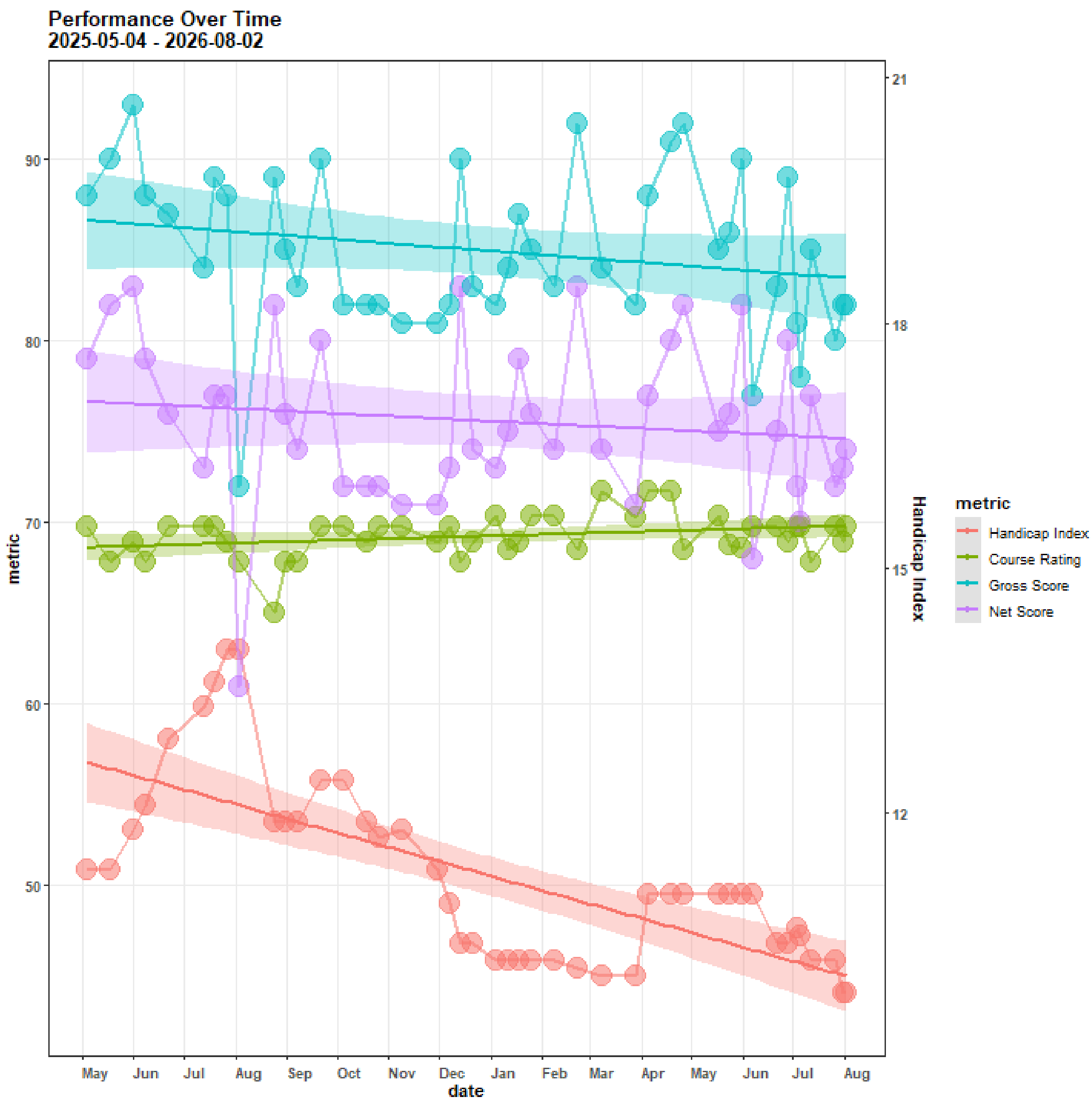
```
## # A tibble: 6 x 10
## # Groups:   date [1]
##   date       club  `rd club strokes` miss_direction `rd club miss dir` rd_avg_yds_to_target rd_avg_y
##   <date>     <chr>             <int> <chr>                       <int>                <dbl>
## 1 2026-08-02 4                     1 short                           1                 220
## 2 2026-08-02 5                     1 short                           1                 210
## 3 2026-08-02 6                     2 long                            1                 200.
## 4 2026-08-02 6                     2 on_target                       1                 200.
## 5 2026-08-02 7                     1 long                            1                 170
## 6 2026-08-02 8                     2 on_target                       2                 162.
```

## Plot Metric Summaries

### Scoring Metrics

```
scoring_metrics |>
  dplyr::mutate(course_rating = dplyr::case_when(grepl(date_course, pattern = 'Sewailo') ~ 68.9, TRUE ~
                `Handicap Index` = `Handicap Index`*4.5) |>
  dplyr::rename(`Course Rating` = course_rating) |>
  dplyr::group_by(date, date_course, `Handicap Index`) |>
  tidyr::pivot_longer(cols = c(`Course Rating`:`Net Score`), names_to = 'metric', values_to = 'value',
  ggplot2::ggplot(aes(x = date_course, y = value)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1,
                          color = factor(metric,
                                         levels = c('Handicap Index', 'Course Rating', 'Gross Score', '
                          fill = factor(metric,
                                        levels = c('Handicap Index', 'Course Rating', 'Gross Score', '
```

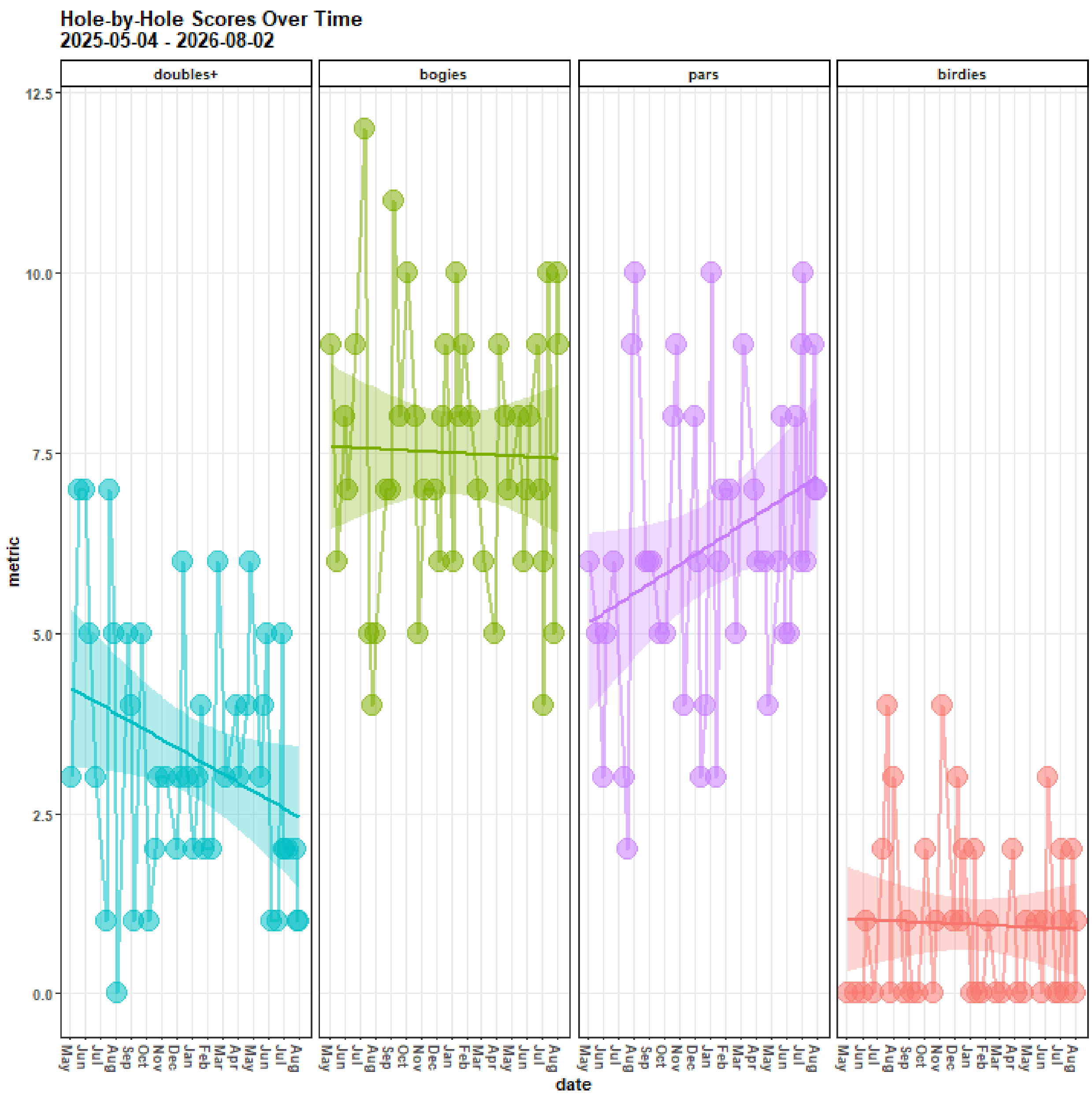
```
ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1,
                        color = factor(metric,
                                       levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
                        fill = factor(metric,
                                      levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
ggplot2::geom_smooth(aes(x = date, y = value, group = metric,
                         color = factor(metric,
                                        levels = c('Handicap Index', 'Course Rating', 'Gross Score',
                         fill = factor(metric,
                                       levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
                     alpha = 0.3, method = 'lm') +
ggplot2::scale_y_continuous(sec.axis = ggplot2::sec_axis(~./4.5, name = 'Handicap Index'))+
ggplot2::labs(title = paste0('Performance Over Time\n',
                             scores |>
                               dplyr::distinct(date) |>
                               dplyr::last() |>
                               unlist() %>% lubridate::as_date(.) |>
                               as.character(),
                             ' - ',
                             scores |>
                               dplyr::distinct(date) |>
                               dplyr::first() |>
                               unlist() %>% lubridate::as_date(.) |>
                               as.character()),
              x = 'date',
              y = 'metric',
              color = 'metric'
              ) +
ggplot2::guides(alpha = 'none', size = 'none', fill = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(panel.grid.minor = ggplot2::element_blank(),
               axis.title = ggplot2::element_text(face = 'bold'),
               axis.text.y = ggplot2::element_text(face = 'bold'),
               axis.text.x = ggplot2::element_text(face = 'bold', angle = 0, hjust = 0, vjust = 0),
               title = ggplot2::element_text(face = 'bold')
               ) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b')
```




### Stroke Metrics

```
stroke_metrics |>
  tidyr::pivot_longer(cols = c(`doubles+`:birdies), names_to = 'metric', values_to = 'value', values_dro
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Hole-by-Hole Scores Over Time\n',
                               scores |>
                                 dplyr::distinct(date) |>
                                 dplyr::last() |>
```

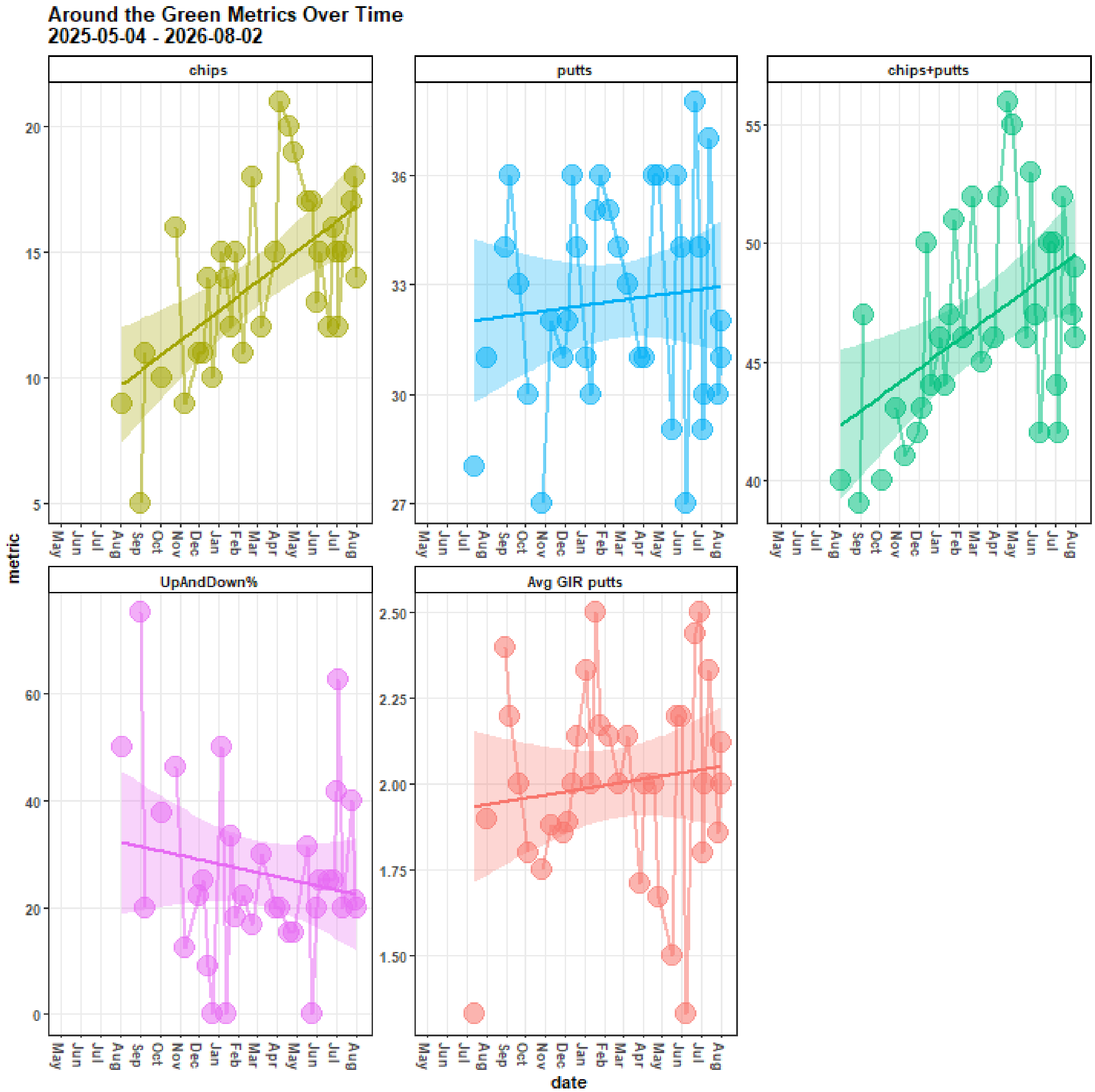
```
                                  unlist() %>% lubridate::as_date(.) |>
                                  as.character(),
                                ' - ',
                                scores |>
                                  dplyr::distinct(date) |>
                                  dplyr::first() |>
                                  unlist() %>% lubridate::as_date(.) |>
                                  as.character()),
                 x = 'date',
                 y = 'metric',
                 color = 'metric', fill = 'metric') +
  ggplot2::guides(alpha = 'none', size = 'none') +
  ggplot2::theme_bw() +
  ggplot2::theme(legend.position = 'none',
                  panel.grid.minor = ggplot2::element_blank(),
                  axis.title = ggplot2::element_text(face = 'bold'),
                  axis.text.y = ggplot2::element_text(face = 'bold'),
                  axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5
                  title = ggplot2::element_text(face = 'bold'),
                  strip.text.x.top = ggplot2::element_text(face = 'bold'),
                  strip.background = ggplot2::element_rect(color = 'black',
                                                           fill = 'white')) +
  ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
  ggplot2::facet_wrap(~factor(metric, levels = c('doubles+', 'bogies', 'pars', 'birdies')), nrow = 1)
```




### Around the Green Metrics

```
atg_metrics |>
  tidyr::pivot_longer(cols = c(chips:`Avg GIR putts`), names_to = 'metric', values_to = 'value', values_
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Around the Green Metrics Over Time\n',
                               scores |>
                                 dplyr::distinct(date) |>
                                 dplyr::last() |>
```

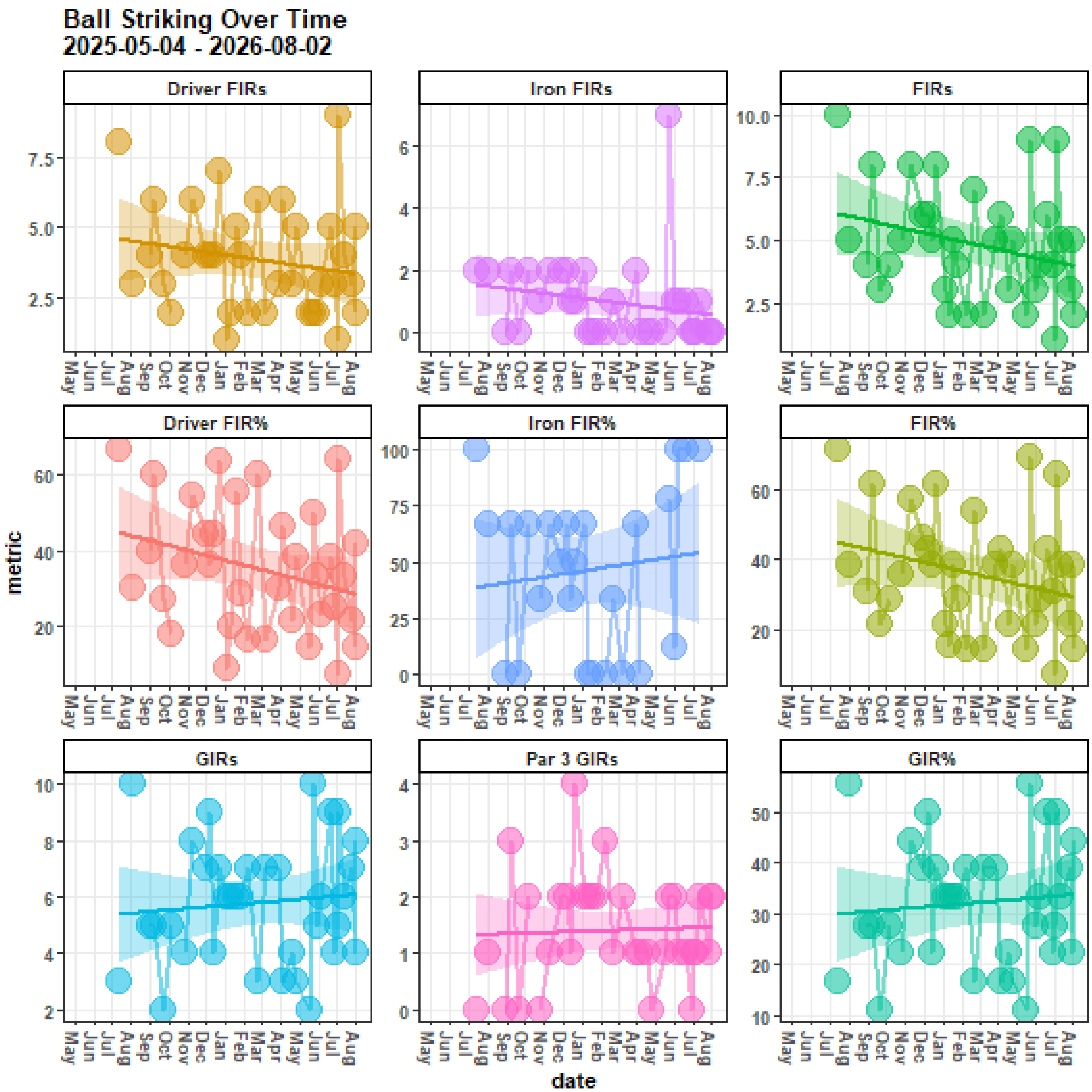
```
                                unlist() %>% lubridate::as_date(.) |>
                                as.character(),
                              ' - ',
                              scores |>
                                dplyr::distinct(date) |>
                                dplyr::first() |>
                                unlist() %>% lubridate::as_date(.) |>
                                as.character()),
                x = 'date',
                y = 'metric',
                color = 'metric', fill = 'metric') +
  ggplot2::guides(alpha = 'none', size = 'none') +
  ggplot2::theme_bw() +
  ggplot2::theme(legend.position = 'none',
                 panel.grid.minor = ggplot2::element_blank(),
                 axis.title = ggplot2::element_text(face = 'bold'),
                 axis.text.y = ggplot2::element_text(face = 'bold'),
                 axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5
                 title = ggplot2::element_text(face = 'bold'),
                 strip.text.x.top = ggplot2::element_text(face = 'bold'),
                 strip.background = ggplot2::element_rect(color = 'black',
                                                          fill = 'white')) +
  ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
  ggplot2::facet_wrap(~factor(metric, levels = c('chips', 'putts', 'chips+putts', 'UpAndDown%', 'Avg GI
```




### Ball Striking Metrics

```
ball_striking_metrics |>
  tidyr::pivot_longer(cols = c(GIRs:`Driver FIR%`), names_to = 'metric', values_to = 'value', values_dr
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Ball Striking Over Time\n',
                               scores |>
                                 dplyr::distinct(date) |>
                                 dplyr::last() |>
```

```
                                  unlist() %>% lubridate::as_date(.) |>
                                  as.character(),
                                ' - ',
                                scores |>
                                  dplyr::distinct(date) |>
                                  dplyr::first() |>
                                  unlist() %>% lubridate::as_date(.) |>
                                  as.character()),
                 x = 'date',
                 y = 'metric',
                 color = 'metric', fill = 'metric') +
  ggplot2::guides(alpha = 'none', size = 'none') +
  ggplot2::theme_bw() +
  ggplot2::theme(legend.position = 'none',
                 panel.grid.minor = ggplot2::element_blank(),
                 axis.title = ggplot2::element_text(face = 'bold'),
                 axis.text.y = ggplot2::element_text(face = 'bold'),
                 axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5)
                 title = ggplot2::element_text(face = 'bold'),
                 strip.text.x.top = ggplot2::element_text(face = 'bold'),
                 strip.background = ggplot2::element_rect(color = 'black',
                                                          fill = 'white')) +
  ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
  ggplot2::facet_wrap(~factor(metric,levels = c('Driver FIRs', 'Iron FIRs', 'FIRs',
                                                 'Driver FIR%', 'Iron FIR%', 'FIR%',
                                                 'GIRs', 'Par 3 GIRs', 'GIR%')), scales = 'free')
```



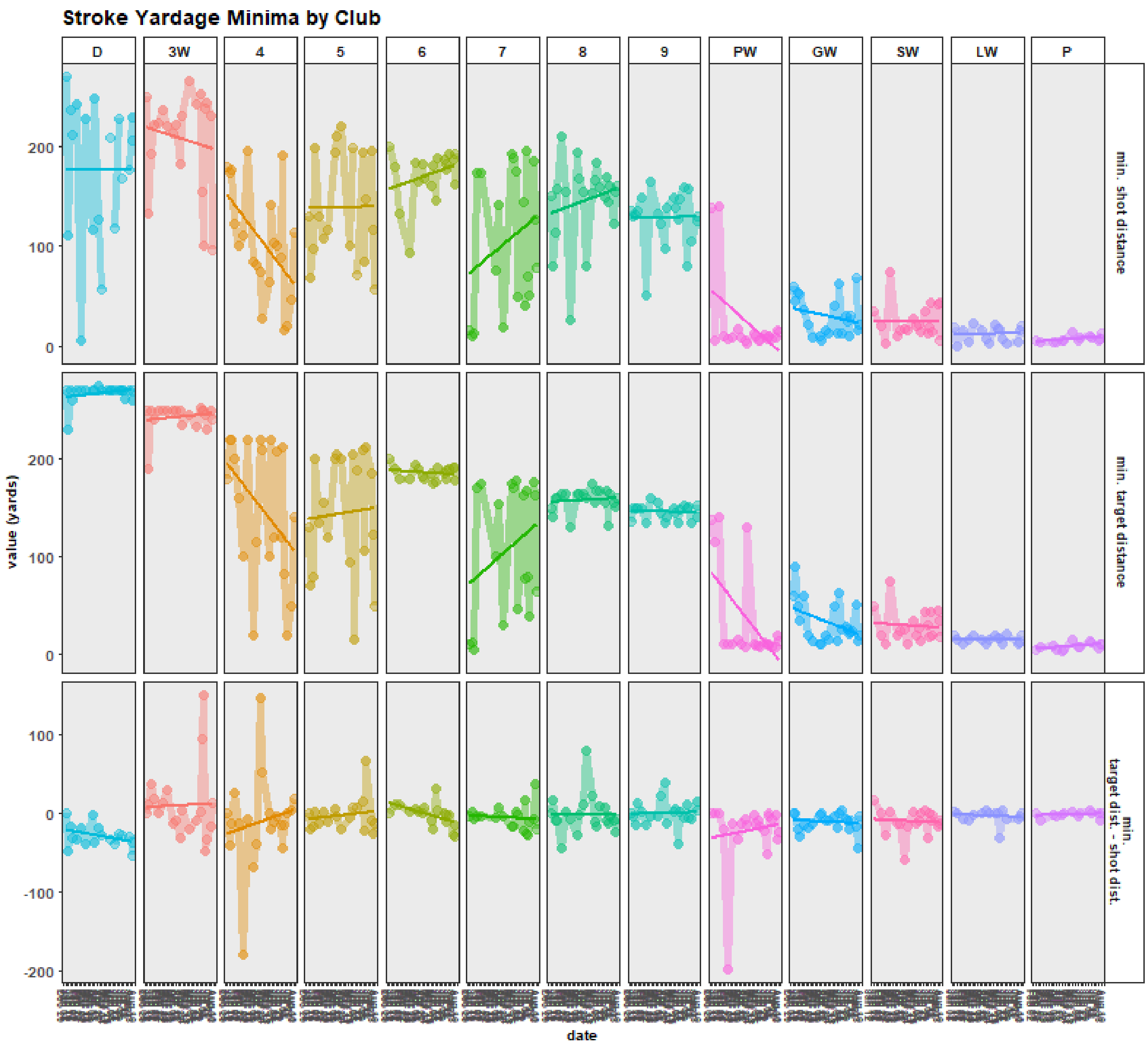

### Stroke Quality Metrics

```
stroke_quality_min |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(c(date:n), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains('yd')), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(grepl(metric, pattern = 'target') ~ 'min. target distance',
                                          grepl(metric, pattern = 'traveled') ~ 'min. shot distance',
                                          grepl(metric, pattern = 'diff') ~ 'min.\ntarget dist. - shot
  dplyr::mutate(date = lubridate::as_date(date),
                club = as.character(club),
```

```
              n = as.integer(n)) |>
  ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
  ggplot2::geom_point(alpha = 0.4, size = 3) +
  ggplot2::geom_line(alpha = 0.4, size = 2) +
  ggplot2::geom_smooth(method = 'lm', se = F) +
  ggplot2::labs(title = 'Stroke Yardage Minima by Club', x = 'date', y = 'value (yards)', fill = 'club'
  ggplot2::guides(alpha = 'none', size = 'none') +
  ggplot2::theme_bw() +
  ggplot2::theme(title = ggplot2::element_text(face = 'bold', size = 12),
               axis.title = ggplot2::element_text(face = 'bold', size = 10),
               axis.text = ggplot2::element_text(face = 'bold', size = 10),
               axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, v
               legend.position = 'none',
               strip.background = ggplot2::element_rect(fill = 'white'),
               strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
               strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
  ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
  ggplot2::facet_grid(

    rows = vars(metric),

    cols = vars(factor(club, levels = c('D', '3W', '4', '5',
                                         '6', '7', '8', '9',
                                         'PW', 'GW', 'SW', 'LW', 'P'))),
                    scales = 'free')
```



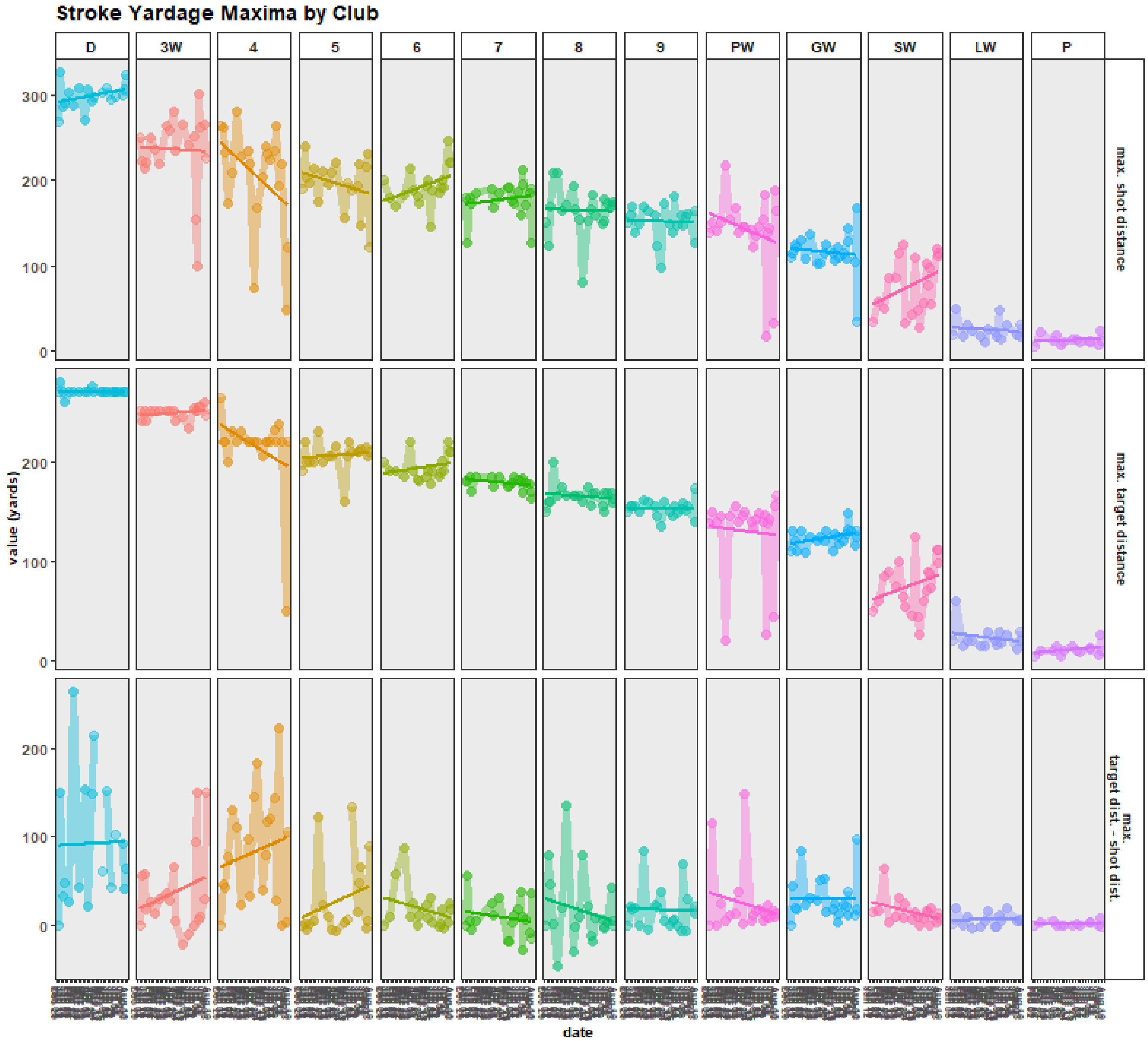

### Minima

```
stroke_quality_max |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(c(date:n), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains('yd')), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(grepl(metric, pattern = 'target') ~ 'max. target distance',
                                          grepl(metric, pattern = 'traveled') ~ 'max. shot distance',
                                          grepl(metric, pattern = 'diff') ~ 'max.\ntarget dist. - shot d
  dplyr::mutate(date = lubridate::as_date(date),
                club = as.character(club),
                n = as.integer(n)) |>
  ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
  ggplot2::geom_point(alpha = 0.4, size = 3) +
  ggplot2::geom_line(alpha = 0.4, size = 2) +
  ggplot2::geom_smooth(method = 'lm', se = F) +
  ggplot2::labs(title = 'Stroke Yardage Maxima by Club', x = 'date', y = 'value (yards)', fill = 'club')
```

```
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(title = ggplot2::element_text(face = 'bold', size = 12),
               axis.title = ggplot2::element_text(face = 'bold', size = 10),
               axis.text = ggplot2::element_text(face = 'bold', size = 10),
               axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, v
               legend.position = 'none',
               strip.background = ggplot2::element_rect(fill = 'white'),
               strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
               strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
ggplot2::facet_grid(

  rows = vars(metric),

  cols = vars(factor(club, levels = c('D', '3W', '4', '5',
                                      '6', '7', '8', '9',
                                      'PW', 'GW', 'SW', 'LW', 'P'))),
                    scales = 'free')
```



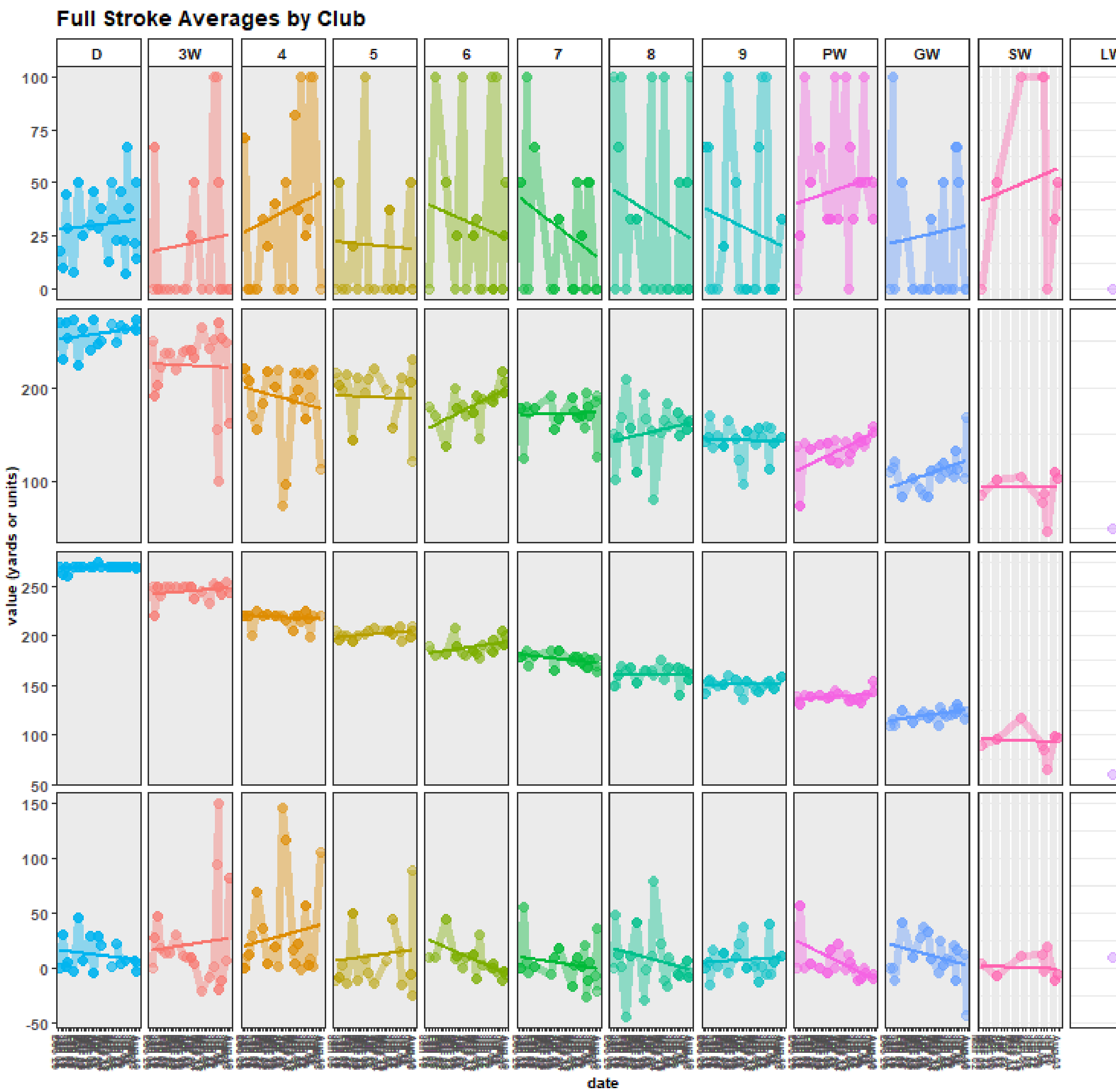

## Maxima

```
full_stroke_quality_avg |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(dplyr::everything(), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains("rd")), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(grepl(metric, pattern = 'target') ~ 'avg. target distance',
                                          grepl(metric, pattern = 'traveled') ~ 'avg. shot distance',
                                          grepl(metric, pattern = 'diff') ~ 'avg.\ntarget dist. - shot d
                                          # grepl(metric, pattern = 'accuracy') ~ 'avg\naccuracy',
                                          grepl(metric, pattern = 'strokes') ~ 'n strokes',
                                          grepl(metric, pattern = 'dir') ~ 'strokes w/club\nin round\nb
                                          grepl(metric, pattern = 'acc') ~ '% accuracy')) |>
  dplyr::filter(!grepl(metric, pattern = 'strokes')) |>
  dplyr::mutate(date = lubridate::as_date(date),
                club = as.character(club),
                vals = as.numeric(vals)
```

```
               ) |>
  ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
  ggplot2::geom_point(alpha = 0.4, size = 3) +
  ggplot2::geom_line(alpha = 0.4, size = 2) +
  ggplot2::geom_smooth(method = 'lm', se = F) +
  ggplot2::labs(title = 'Full Stroke Averages by Club', x = 'date', y = 'value (yards or units)', fill =
  ggplot2::guides(alpha = 'none', size = 'none') +
  ggplot2::theme_bw() +
  ggplot2::theme(title = ggplot2::element_text(face = 'bold', size = 12),
                 axis.title = ggplot2::element_text(face = 'bold', size = 10),
                 axis.text = ggplot2::element_text(face = 'bold', size = 10),
                 axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, v
                 legend.position = 'none',
                 strip.background = ggplot2::element_rect(fill = 'white'),
                 strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
                 strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
  ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
  ggplot2::facet_grid(

    rows = vars(metric),

    cols = vars(factor(club, levels = c('D', '3W', '4', '5',
                                        '6', '7', '8', '9',
                                        'PW', 'GW', 'SW', 'LW', 'P'))),
                    scales = 'free')
```

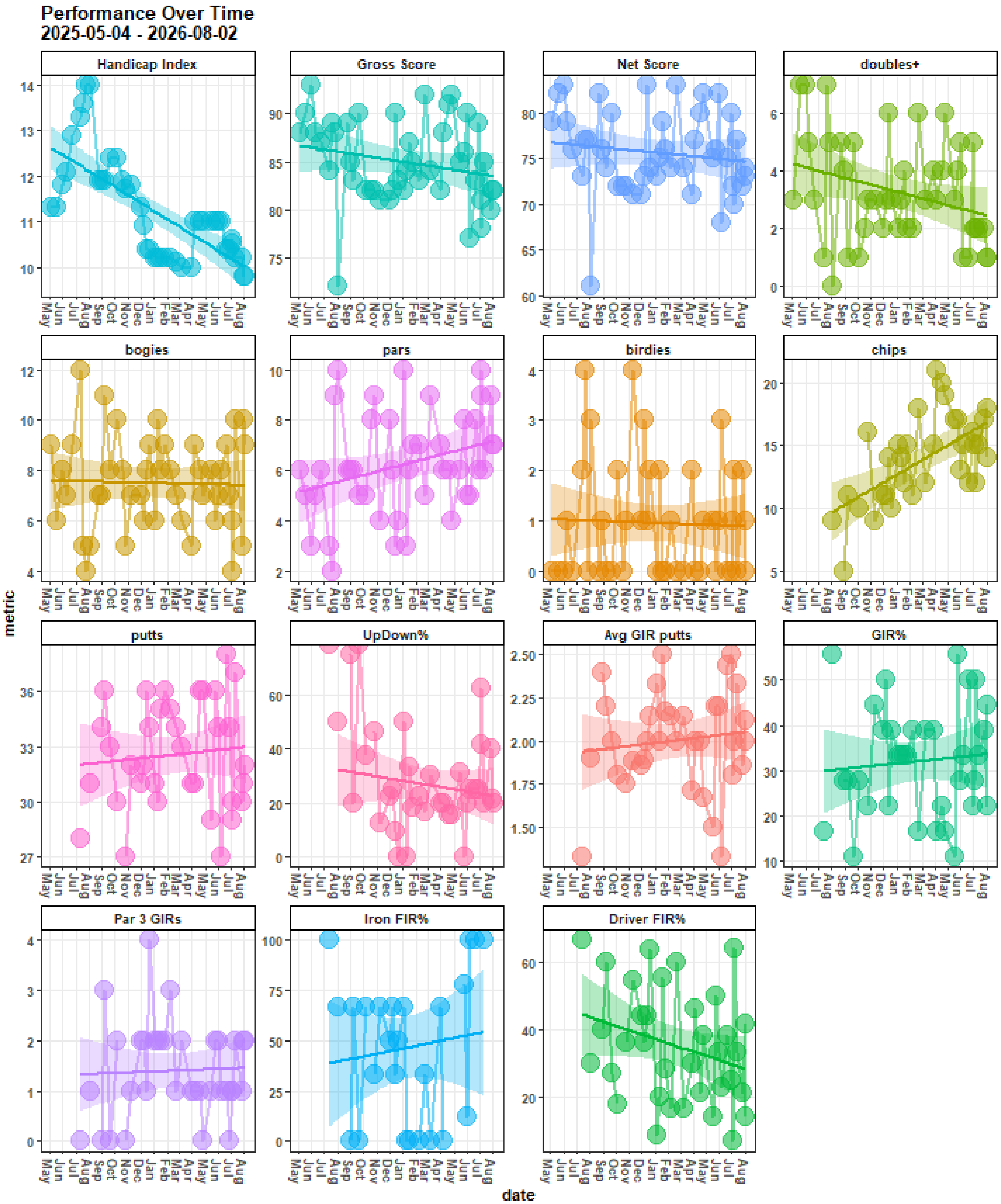



### Full Stroke Average

### Main Metrics

```
scores_sum |>
  dplyr::mutate(course_rating = dplyr::case_when(grepl(date_course, pattern = 'Sewailo') ~ 68.9, TRUE ~
  tidyr::pivot_longer(cols = c(`Handicap Index`, `Gross Score`, `Net Score`,
                               `doubles+`, bogies, pars, birdies, chips, putts,
                               `UpDown%`,`Avg GIR putts`, `GIR%`, `Par 3 GIRs`,
                               `Iron FIR%`, `Driver FIR%`), names_to = 'metric', values_to = 'value', va
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Performance Over Time\n',
                               scores |>
                                 dplyr::distinct(date) |>
                                 dplyr::last() |>
```

```
                                  unlist() %>% lubridate::as_date(.) |>
                                  as.character(),
                                ' - ',
                                scores |>
                                  dplyr::distinct(date) |>
                                  dplyr::first() |>
                                  unlist() %>% lubridate::as_date(.) |>
                                  as.character()),
               x = 'date',
               y = 'metric',
               color = 'metric', fill = 'metric') +
  ggplot2::guides(alpha = 'none', size = 'none') +
  ggplot2::theme_bw() +
  ggplot2::theme(legend.position = 'none',
                panel.grid.minor = ggplot2::element_blank(),
                axis.title = ggplot2::element_text(face = 'bold'),
                axis.text.y = ggplot2::element_text(face = 'bold'),
                axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5
                title = ggplot2::element_text(face = 'bold'),
                strip.text.x.top = ggplot2::element_text(face = 'bold'),
                strip.background = ggplot2::element_rect(color = 'black',
                                                         fill = 'white')) +
  ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
  ggplot2::facet_wrap(~factor(metric,
                     levels = c('Handicap Index', 'Gross Score', 'Net Score',
                                'doubles+', 'bogies', 'pars', 'birdies',
                                'chips', 'putts', 'UpDown%', 'Avg GIR putts',
                                'GIR%', 'Par 3 GIRs', 'Iron FIR%', 'Driver FIR%')),
                     scales = 'free',
                     ncol = 4)
```


Performance Over Time
2025-05-04 - 2026-08-02
Handicap Index
Gross Score
Net Score
doubles+
bogies
pars
birdies
chips
putts
UpDown%
Avg GIR putts
GIR%
Par 3 GIRs
Iron FIR%
Driver FIR%
metric
date